Person Detection in Movie Posters to Understand Popularity

Group 1: Alison Miller (leader),
Stephanie Diau, Geethan Sundaram
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Project Details

Motivation:	The power of an ensemble cast has risen in series on streaming platforms, is the ensemble becoming the leader in the theater too?
Hypothesis:	There exists a statistically significant positive relationship between the amount of person in a movie poster and popularity score of that movie in the past 10 year period.
Research Question:	Is there a correlation between the popularity of a movie and the amount of individuals identified in the movie's poster?
Modeling Approach:	Poster Set > Preprocess > Yolo v8 > Yolo v8 (L) > Best-Fit Model > Evaluate
Goal:	This project aims to understand if the # of persons identified through detection in a movie poster correlates with its popularity (in 2013-2023).

Data Explanation & Acquisition

★ Acquisition: We obtained our dataset from “movie_posters_100k_controlnet” set on Hugging Face.

★ Dataset Size:

- Original: 91,000 rows x 17 columns
- Cleaned: 10,000 x 6 columns
  - Reduced to last 10 years of data.
  - Kept only rows with image_height = 750. (>90% with category)
  - Dropped non-relevant columns.
  - Created images folder and poster path in metadata .csv.
- New Dataset: 10,000 rows x 8 columns
  - Detected persons columns.
    - Custom (UL) or non-custom (L).

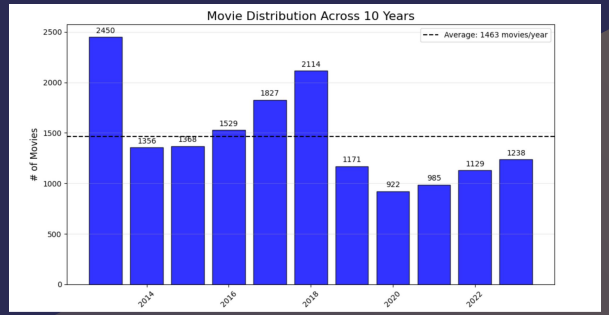
	image_path	title	popularity	release_date	image_height	image_width
1	red_posters/poster_0.jpg	57 Seconds	1036.051	2023-09-29	750	500
2	red_posters/poster_1.jpg	ead Reckoning Part One	3132.329	2023-07-08	750	500
3	red_posters/poster_2.jpg	Expend4bles	1063.874	2023-09-15	750	500
4	red_posters/poster_3.jpg	Sound of Freedom	1271.779	2023-07-03	750	500
5	red_posters/poster_4.jpg	Nowhere	1290.251	2023-09-29	750	500
6	red_posters/poster_5.jpg	The Nun II	1706.801	2023-09-06	750	500
7	red_posters/poster_6.jpg	Talk to Me	1188.455	2023-07-26	750	500
8	red_posters/poster_7.jpg	Gran Turismo	1025.362	2023-08-09	750	500
9	red_posters/poster_8.jpg	Ballerina	1091.044	2023-10-05	750	500
10	red_posters/poster_9.jpg	Retribution	697.448	2023-08-23	750	500
11	ed_posters/poster_10.jpg	The Kill Room	705.024	2023-09-14	750	500
12	ed_posters/poster_11.jpg	Muzzle	855.615	2023-09-29	750	500
13	ed_posters/poster_12.jpg	Meg 2: The Trench	942.047	2023-08-02	750	500
14	ed_posters/poster_13.jpg	Fast X	761.431	2023-05-17	750	500
15	ed_posters/poster_14.jpg	Patrol: The Mighty Movie	897.568	2023-09-21	750	500
16	ed_posters/poster_15.jpg	After Everything	667.278	2023-09-13	750	500
17	ed_posters/poster_16.jpg	Barbie	637.727	2023-07-19	750	500
18	ed_posters/poster_17.jpg	Overhaul	579.324	2023-09-27	750	500
19	ed_posters/poster_18.jpg	Uri: The Surgical Strike	532.808	2019-01-11	750	500
20	ed_posters/poster_19.jpg	Elemental	558.946	2023-06-14	750	500
21	ed_posters/poster_20.jpg	rmers: Rise of the Beasts	546.911	2023-06-06	750	500
22	ed_posters/poster_21.jpg	Saw X	521.01	2023-09-26	750	500
23	ed_posters/poster_22.jpg	Totally Killer	490.274	2023-09-28	750	500
24	ed_posters/poster_23.jpg	Haunted Mansion	486.811	2023-07-26	750	500

Poster Metadata .csv

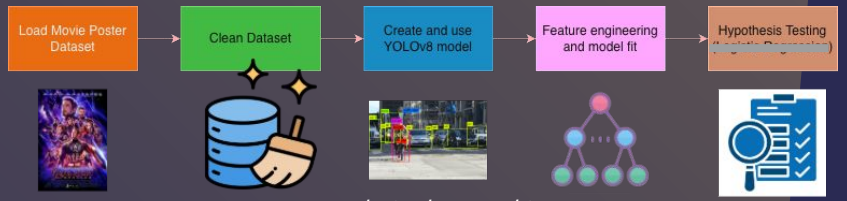
Poster Image Example

Analysis Plan & Justification

- Preprocessing Data:
  - Streaming for download.
  - 91k rows, 7 col -> 10k rows, 6 col.
    - Counts added after Yolo.
- Creating Yolo v.8:
  - Detection with pre trained class.
  - Detection with custom class.
    - Hand label subset.
- Modeling:
  - Identify what is suited:
    - Gradient Boost.
    - Regression Model.
- Evaluate Results:
  - Create Visualizations.
  - Identify any significance.



Initial EDA : Visualize spread and effect of other kept columns (budget, revenue) on popularity.



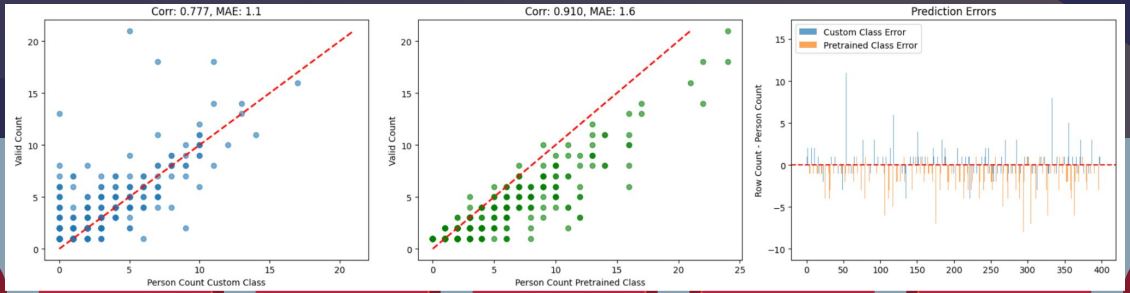
Analysis Plan Graphic

Tricky Analysis Decision

Before modeling we had to make the choice between our Yolov8s iterations, custom or pre trained class, for gradient boost/modeling.

- ★ Evaluated both models on the same set of 400 images for detection accuracy compared to literal counted persons with standard confidence threshold.
- ★ Pre trained: Higher correlation, higher MAE.
- ★ Custom: Lower corr, lower MAE.

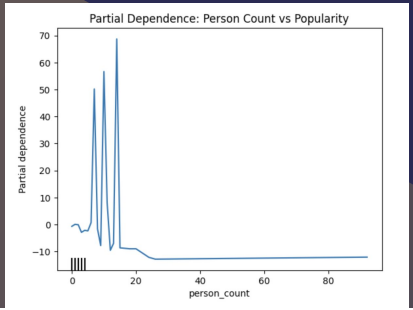
Ended up choosing custom class given lower MAE. However we acknowledge that instead choosing the pre trained could have given us a different outlook on further modeling. The custom and pre trained had their prediction error through opposite methods.



Bias and Uncertainty

- Bias from original dataset:
  - Though dataset is large, does not guarantee full scope of industry. Did best to find set with variability: international, budget size, genre.
- Detection:
  - Animation, crowds, and personified non-human characters may present struggles.
  - Opted into custom training, hand bounding boxes to include these characters in subset.
- General ambiguity of poster designs.

Results & Conclusion

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Gradient Boosting MSE: 4524.5964981271245
Gradient Boosting RMSE: 67.26512096270343
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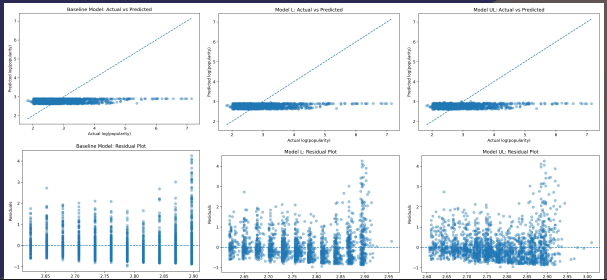
Low counts (0–5): no effect
Mid counts (5–15): noisy / unreliable
High counts (>20): no data → meaningless

High RMSE



	Model	Test_RMSE	Test_MAE	Test_R2	OLS_R2	OLS_Adj_R2	AIC	BIC
0	Baseline: release_year	0.682316	0.479348	0.017557	0.016450	0.016351	19653.064965	19667.485646
1	Model L: person_count_L + release_year	0.682146	0.479295	0.018047	0.016658	0.016462	19652.944446	19674.575467
2	Model UL: person_count_UL + release_year	0.681476	0.478765	0.019974	0.017599	0.017402	19643.377288	19665.008309

Baseline and unlabeled: statistically significant!
Labeled: positive relationship, not SS
Very low RMSE and MAE!!

But also very low R^2 values

CONCLUSION: Across both linear regression and gradient boosting models, we find no meaningful evidence that the number of people in a movie poster is an important predictor of movie popularity

Next Steps

1. Different feature

Placement of people, lead actor, facial expressions

2. Use more variables

Budget, genre, cast popularity

3. Different model

Interaction terms, polynomial

References

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Questions?